

FlavorForge – An Interactive Digital Recipe Book

Web Technologies (ICT 1209) Mini Project Report

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1. PROJECT OVERVIEW

1.1 Purpose

The primary objective of this mini project is to develop an **Interactive Digital Recipe Book** named **FlavorForge**. Traditional paper cookbooks can be hard to organize, search through, and update. This web application provides a clean, user friendly digital platform where users can easily search for recipes, filter them based on categories or cooking duration, and submit their own custom recipes.

1.2 Target Audience

- **Home Cooks and Food Enthusiasts:** People looking for a convenient digital space to browse and organize daily recipes.
- **Students and Beginners:** Novice cooks who need quick recipe ideas with step by step preparation steps and ingredient lists.

1.3 Key Features

- **Interactive Search and Filtering:** Users can filter recipes by categories (such as Breakfast, Lunch, Dinner, Dessert), search by keywords, and filter by cooking time.
- **User Account Management:** Users can register for a new account, log in securely, and log out.
- **Recipe Submission (CRUD):** Logged in users can post their own recipes with ingredients, preparation steps, and image links.
- **Recipe Detail Popup (Modal):** Clicking on any recipe opens a popup window displaying full ingredients and step by step instructions.
- **Contact Us Module:** Visitors can send inquiries or feedback through a contact form, which saves the message to the database.

2. SYSTEM DESIGN

2.1 Database Schema

The database for this project is named `recipe_book` and contains three relational tables: users, recipes, and messages.

1. users Table

Stores registered user account credentials.

Field	Type	Null	Key	Default	Description
id	INT(11)	NO	PRI	NULL	Auto-increment primary key
username	VARCHAR(50)	NO	UNI	NULL	Unique username
email	VARCHAR(100)	NO	UNI	NULL	Unique email address
password	VARCHAR(255)	NO		NULL	Encrypted password (BCRYPT)
created_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	Account creation timestamp

2. messages Table

Stores inquiries submitted through the contact page.

Field	Type	Null	Key	Default	Description
id	INT(11)	NO	PRI	NULL	Auto-increment primary key
name	VARCHAR(100)	NO		NULL	Sender's full name
email	VARCHAR(100)	NO		NULL	Sender's email address
message	TEXT	NO		NULL	Message content
created_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	Submission timestamp

3. recipes Table

Stores all recipe records created by users.

Field	Type	Null	Key	Default	Description
id	INT(11)	NO	PRI	NULL	Auto-increment primary key
title	VARCHAR(150)	NO		NULL	Recipe title
category	VARCHAR(50)	NO		NULL	Category (e.g. Breakfast, Lunch)
prep_time	INT(11)	NO		NULL	Cooking time in minutes
difficulty	VARCHAR(20)	NO		NULL	Difficulty level (Easy, Medium, Hard)
ingredients	TEXT	NO		NULL	List of ingredients
instructions	TEXT	NO		NULL	Step-by-step instructions
image_url	VARCHAR(255)	YES		NULL	Image path or web link
user_id	INT(11)	YES	MUL	NULL	Foreign Key referencing users(id)
created_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	Recipe creation timestamp

2.2 Folder Structure

```
Digital Recipe Book/
├── auth/
│   ├── login.php
│   ├── logout.php
│   └── register.php
├── css/
│   └── style.css
├── includes/
│   ├── db.php
│   └── functions.php
├── js/
│   ├── filter.js
│   ├── main.js
│   └── validation.js
├── pictures/
├── add_recipe.php
├── contact.php
├── dashboard.php
├── database.sql
├── index.php
└── README.md
```

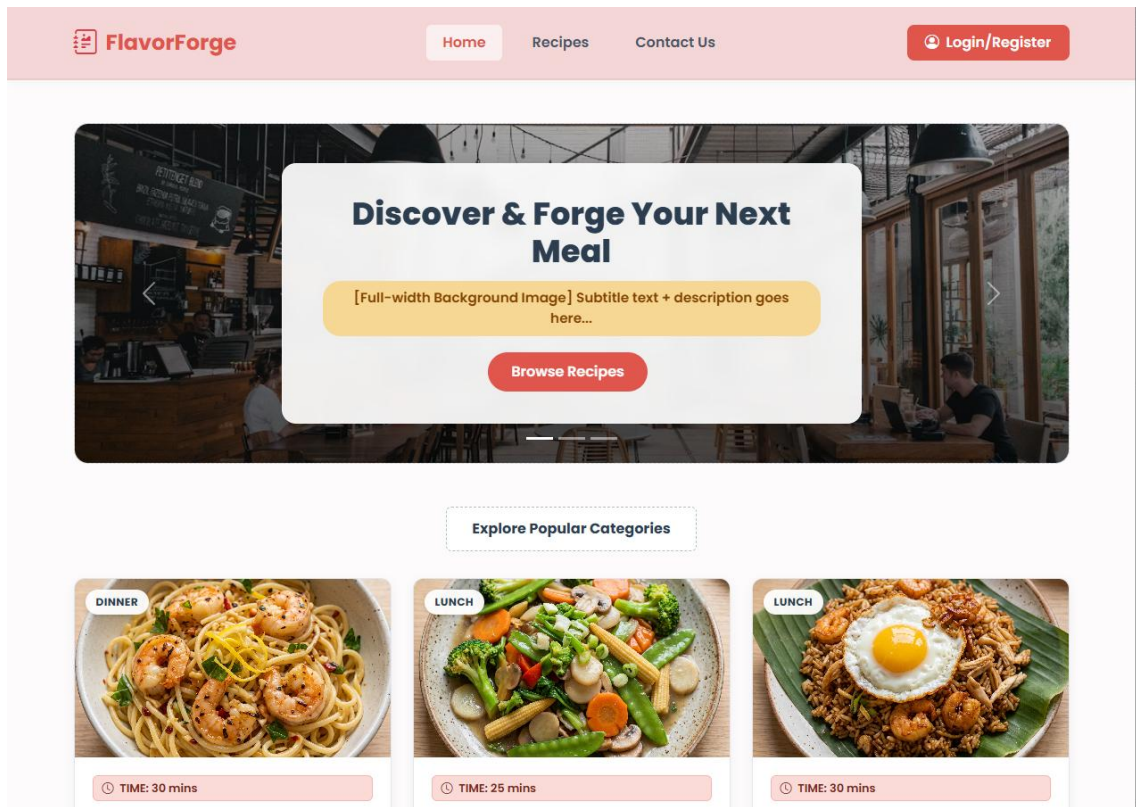
3. IMPLEMENTATION

3.1 Homepage and Dynamic Recipe Display

The homepage (index.php) serves as the main landing view where users can explore active recipes fetched directly from the MySQL database using PHP PDO queries.

- **Featured Hero Carousel:** Displays highlighted recipes with call to action buttons.
- **Dynamic Grid Cards:** Displays recipe cards showing title, cooking duration, difficulty badge, and meal category.

Figure 3.1: Homepage Layout & Dynamic Recipe Grid

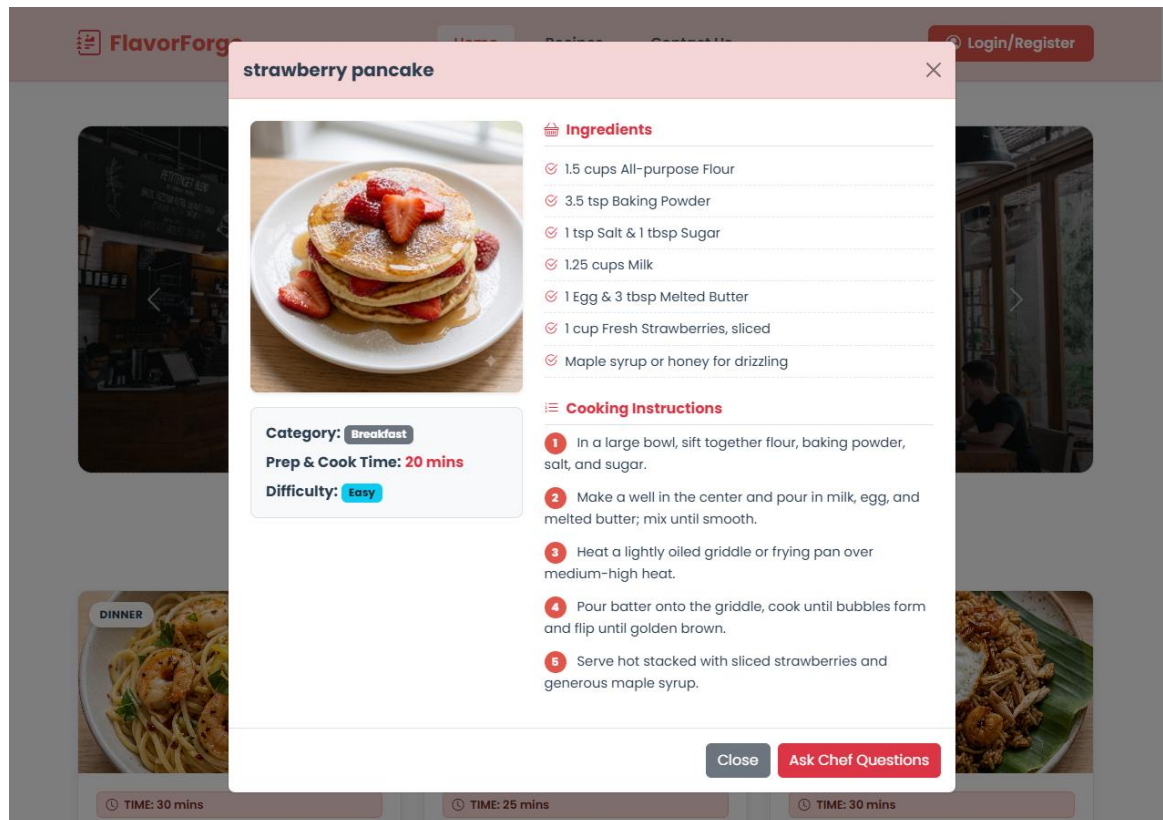


3.2 Real-Time Recipe Search and Category Filtering

The filtering engine (filter.js) allows users to search by recipe names or filter by categories without reloading the page.

- **Category Filter Buttons:** Quick filters for Breakfast, Lunch, Dinner, and Desserts.
- **Keyword Search Bar:** Real time DOM filtering based on text input.

Figure 3.2: Real-Time Category Search and Filtering

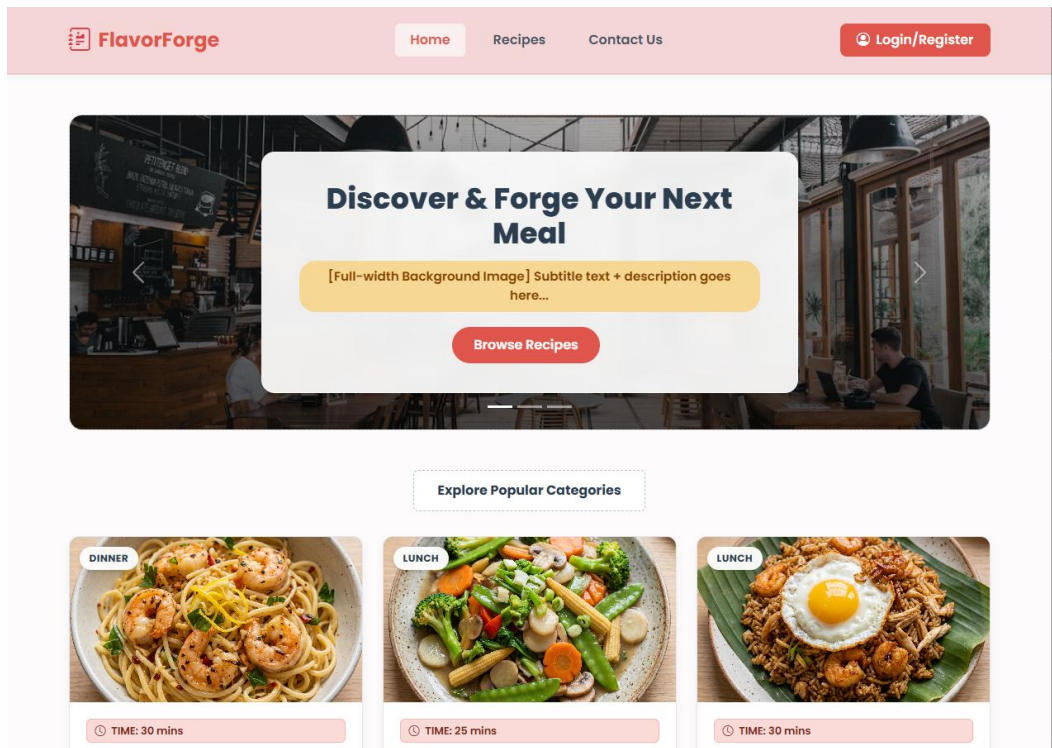


3.3 Interactive Recipe Detail Popup Window

Clicking on any recipe card triggers a Bootstrap modal popup window (main.js).

- The modal dynamically populates the full ingredient checklist, cooking preparation steps, prep time, and recipe image.
- PHP converts recipe records into JSON format (window.dbRecipes) to enable instant dynamic modal rendering.

Figure 3.3: Interactive Recipe Detail Modal Window



3.4 User Authentication System

The user authentication module manages secure user registration, login, and session tracking.

- **User Registration (auth/register.php):** Validates input fields, checks for existing usernames/emails, and securely encrypts passwords using `password_hash(PASSWORD_BCRYPT)`.
- **User Login (auth/login.php):** Verifies credentials using `password_verify()`, regenerates session IDs (`session_regenerate_id(true)`), and starts a logged-in session.
- **User Logout (auth/logout.php):** Clears session arrays and destroys active login sessions safely.

Figure 3.4: User Login and Registration Interface Forms

The image displays two mobile application interface forms for FlavorForge. The left form, titled 'Welcome Back', is for user login and includes fields for 'Email Address' (with a placeholder 'user@example.com') and 'Password' (with a masked input '*****'). It features a red 'Login' button and a link to 'Register here' for users without an account. The right form, titled 'Create Account', is for user registration and includes fields for 'Username' (placeholder 'e.g. chef_sithumi'), 'Email Address' (placeholder 'user@example.com'), 'Password' (placeholder 'At least 6 characters'), and 'Confirm Password' (placeholder 'Re-enter password'). It features a red 'Create Account' button and a link to 'Login here' for users who already have an account. Both forms are set against a light gray background with the FlavorForge logo at the top.

3.5 Recipe Submission and Dashboard

Logged-in users gain access to the user dashboard (dashboard.php) where they can submit new recipes via add_recipe.php.

- Includes input fields for recipe title, category, preparation time, difficulty level, multiline ingredients, step by step instructions, and image file paths.
- Submissions are validated on both client side and server side before storing into the MySQL database.

Figure 3.5: User Dashboard & Recipe Submission Form

The image shows a screenshot of a web application's user dashboard with a modal form titled 'Add New Recipe (MySQL CRUD Module)'. The form contains several input fields: 'Recipe Title' (placeholder 'e.g. Creamy Mushroom Pasta'), 'Category' (dropdown menu with 'Dinner' selected), 'Cooking Time (minutes)' (placeholder 'e.g. 30'), 'Difficulty Level' (dropdown menu with 'Medium' selected), 'Image URL (Optional)' (placeholder 'https://images.unsplash.com/...'), 'Ingredients (One per line)' (text area with '2 cups Flour', '1 tsp Salt', and '1 cup Water'), and 'Cooking Instructions (Step-by-step)' (text area with '1. Mix ingredients in a bowl...' and '2. Bake at 180°C for 25 minutes...'). At the bottom of the form are 'Cancel' and 'Save Recipe to MySQL' buttons. The background shows a blurred view of the dashboard with navigation links like 'Home', 'Recipes', and 'Contact Us', and a sidebar with filters.

3.6 Data-Driven Contact Page

The contact page (contact.php) provides an inquiry form for users to send feedback.

- Form inputs are validated and saved into the messages table in the database using PDO prepared statements.
- Displays feedback success/error alert banners after submission.

Figure 3.6: Data-Driven Contact Page and Inquiry Form

FlavorForge Home Recipes **Contact Us** Login/Register

Contact us

We would love to hear from you

Get in Touch

Location
Rajarata University of Sri Lanka,
Anuradhapura,
Mihintale

Email
flavorforge@gmail.com
info@flavorforge.com

Hours
Mon - Fri: 9:00 AM - 5:00 PM
Sat - Sun: Closed

For urgent inquiries, please email directly and include "URGENT" in the subject line.

Send us Message

Full Name
e.g. Sithumi Perera

Email Address
e.g. sithumi@gmail.com

Your Message
Type your message here...

Submit Message

4. CHALLENGES AND SOLUTIONS

4.1 Challenge 1: Foreign Key Order in SQL Re-Import

Problem: When re importing database.sql, dropping the users table failed because the recipes table held an active foreign key constraint referencing users(id).

Solution: Added foreign key check flags (SET FOREIGN_KEY_CHECKS = 0; at the top and SET FOREIGN_KEY_CHECKS = 1; at the bottom) in database.sql to allow smooth table recreation without drop order errors.

4.2 Challenge 2: Modal Popup Not Displaying Correct Recipe Data

Problem: Newly added recipes from the database did not show their instructions or ingredients inside the recipe detail modal window.

Solution: Encoded recipe data from PHP into a global JSON array (window.dbRecipes) and updated main.js to dynamically read JSON data when the show.bs.modal event triggers.

4.3 Challenge 3: Prevention of SQL Injection and XSS Vulnerabilities

Problem: Malicious user inputs could lead to SQL injection or Cross Site Scripting (XSS) attacks.

Solution: Used PHP Data Objects (PDO) with parameterized prepared statements for all database queries, and applied htmlspecialchars() to sanitize all text before outputting to HTML pages.

5. INDIVIDUAL CONTRIBUTION

5.1 Contribution of N.H Jayasinghe (Reg No: ITT/2024/049 | Index No: 2734)

- Designed the relational database schema and created table structures for users, recipes, and messages in SQL.
- Wrote the database connection script using PHP PDO and helper utility functions for string sanitization.
- Developed the backend logic for recipe creation (add_recipe.php) including input parsing and SQL insert queries.
- Configured sample dataset scripts and mapped default recipe images in the pictures folder.
- Prepared the technical documentation and project report.

5.2 Contribution of M.S.P Perera (Reg No: ITT/2024/082 | Index No: 2767)

- Designed and built the responsive frontend templates for the homepage, user dashboard, contact page, and login/register forms using Bootstrap 5.
- Developed the user authentication system, including user registration with password hashing, login controller with session handling, and logout functionality.
- Implemented JavaScript logic for real-time recipe category filtering, keyword searching, and client-side form validation.
- Created the interactive recipe detail modal popup handler in JavaScript.
- Handled the contact form submission logic to insert messages into the database.

6. SETUP INSTRUCTIONS

6.1 Prerequisites

- **XAMPP Control Panel** (or any local web server stack containing Apache PHP 8+ and MySQL/MariaDB).

6.2 Database Import Steps

1. Launch **XAMPP Control Panel** and start both **Apache** and **MySQL** services.
2. Open a web browser and navigate to `http://localhost/phpmyadmin/`.
3. Create a new database named `recipe_book` (or select the **Import** tab).
4. Click **Choose File** and select `C:\xampp\htdocs\Digital Recipe Book\database.sql`.
5. Click **Go** at the bottom to execute the SQL script. This initializes tables (users, messages, recipes) and seeds initial sample recipes.

6.3 Running the Application

1. Place the project directory inside the XAMPP web root folder:
`C:\xampp\htdocs\Digital Recipe Book\`
2. Open your browser and type the URL:
`http://localhost/Digital%20Recipe%20Book/index.php`
3. Test credentials for logged-in features:
 - **Username:** `chef_jayasinghe` | **Password:** `password123`
 - **Username:** `perera_cook` | **Password:** `password123`

7. REFERENCES

1. PHP Manual, *"PHP Data Objects (PDO) and Security,"* PHP Documentation Group, 2026. Available: <https://www.php.net/manual/en/book.pdo.php>
2. Bootstrap 5 Documentation, *"Components, Grid System, and Modals,"* Bootstrap Team, 2026. Available: <https://getbootstrap.com/docs/5.3/>
3. Mozilla Developer Network (MDN), *"JavaScript Guide and DOM Events,"* 2026. Available: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
4. MySQL Reference Manual, *"MySQL 8.0 Relational Tables and Constraints,"* Oracle Corporation, 2026.
5. W3Schools, *"PHP & MySQL Web Development Tutorials,"* 2026.